

Impact of Baseline Hard Drug Use on 2-Year Treatment Response in HIV-Infected Men

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February 18, 2026

Research Question

Population: 715 HIV-infected men initiating HAART.

Objective: Assess whether 2-year outcomes improve and whether baseline hard drug use modifies these changes.

Outcomes (continuous, change from baseline to 2 years):

- ▶ Viral load (VLOAD)
- ▶ CD4⁺ T cell count (LEU3N)
- ▶ Physical health score (AGG_PHYS)
- ▶ Mental health score (AGG_MENT)

Exposure: Baseline hard drug use (yes/no)

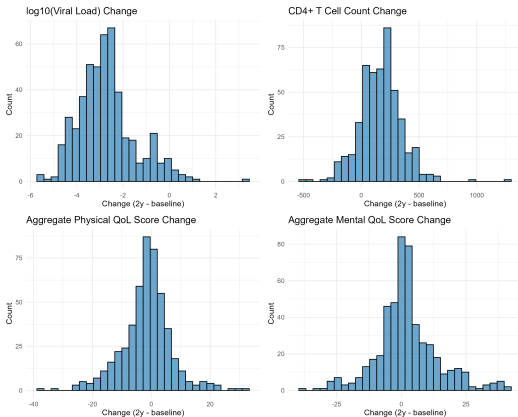
Confounders: Age, race/ethnicity, education, BMI, depression, alcohol, smoking

Data Overview

Sample: 506 individuals with both baseline and 2-year data (Non-users: 467; Users: 39).

Sample size by outcome: Viral load and CD4: 461; Physical and mental health: 470.

Data processing: Removed outliers for CD4 and BMI; re-coded race and education; $\log_{10}$ transform for viral load.



Analysis Plan

Exploratory Analysis:

- ▶ t-tests comparing outcomes between drug users vs non-users
- ▶ Examine missingness (MCAR) and consider imputation

Regression Analysis:

- ▶ Linear regression for each outcome
- ▶ Frequentist: estimate, 95% CI, p-value

Bayesian Analysis:

- ▶ Stan with weakly informative priors
- ▶ Check trace plots, R-hat, ESS
- ▶ Posterior P-value for meaningful changes

Model Selection:

- ▶ Backward selection: AIC (frequentist), WAIC (Bayesian)
- ▶ Compare results across methods